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Relationship between Time Since the Last Fire and Flowering in *Telopea speciosissima* R. Br. and *Lambertia formosa* Sm.

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Abstract

For *Telopea speciosissima* the percentage (or density) of plants in flower, the average number of flowers per inflorescence and the average number of fruits per inflorescence all appear to peak 2 years after a summer fire and decline thereafter. On the other hand, the average height of the inflorescences above the ground increases with time since the last fire. For *Lambertia formosa* the density of inflorescences appears to reach a peak 2 or 3 years after a summer fire.

Introduction

The effects of fire on the Australian flora are many and are increasingly well studied (see Gill *et al.* 1981). There appears, however, to be little quantitative information on the relationships between time since the last fire and flowering of Australian plants (Gill 1981; Specht 1981). Only for *Xanthorrhoea australis* is there a well established relationship (Specht *et al.* 1958; Garnet 1971; Gill and Ingwersen 1976; Gill 1981): in some places this species flowers only rarely in the absence of fire (Specht *et al.* 1958; Garnet 1971).

The aim of this paper is to analyse spatial and temporal variation in flowering of *Telopea speciosissima* (waratah) and *Lambertia formosa* with respect to time since the last fire. This analysis is possible as a result of studies on the pollination biology of these plants and the foraging ecology of honeyeaters that visit them for nectar (Pyke 1980, 1981*a*, 1981*b*).

Methods

The data presented were collected between December 1979 and December 1981 in the Brisbane Water National Park (lat. 33°32′22″S., long. 151°16′56″E.; elevation 170 m above sea level) and Ku-ring-gai Chase National Park (lat. 33°36′18″S., long. 151°16′16″E.; elevation 145 m above sea level). In each region, study sites are located in plateau-type, low open-forest (Benson and Fallding 1981) dominated by *Eucalyptus gummifera* (Gaertn.) Hochr., *E. haemastoma* Sm. and *E. oblonga* DC., with a shrub understory composed mainly of *Lambertia formosa* Sm. Both regions are located on plateau tops and hence all study sites have the same aspect. In both regions soils are deep, yellow and earthy (Benson and Fallding 1981) with no apparent differences between the study sites.

The Brisbane Water study site contains two areas in terms of fire history: one area last burnt in December 1976, the other in December 1979. The two areas are separated by a vehicle trail which stopped the more recent fire from spreading further. There are no apparent differences between the two areas in terms of soil or vegetation. The entire Ku-ring-gai Chase study site last burnt in 1965.

Flowering in *Telopea speciosissima* has been recorded in two ways. In the 1976 burn study area in Brisbane Water National Park, the total number of flowering plants was recorded for the spring flowering periods of 1979, 1980 and 1981. For each inflorescence

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the total number of flowers, height above the ground and number of fruits were also recorded. In each year peak flowering occurred about mid September with high synchrony among plants. In this area and in the other two study areas, permanent circular plots of 20 m radius were established and the numbers of flowering plants within these circles counted every 3 weeks during spring 1981. The numbers of non-flowering plants in each circle were also recorded for spring 1981. Flowering in *Lambertia formosa* was recorded by counting the number of inflorescences in each circle during spring 1981 and in two of the present Brisbane Water circles during spring 1980. During spring 1981, there were four circles in the Brisbane Water area of the 1976 fire, two in the area of the 1979 fire and two in the Ku-ring-gai Chase area.

Results

Flowering of Telopea *in Successive Years*

In the Brisbane Water area which last burnt in December 1976, the total number of flowering *Telopea* plants has declined dramatically since spring 1979 (Table 1). Since spring

Table 1. Flowering of *Telopea speciosissima* in areas of 1976 and 1979 fires during successive years

Area	Flowering time	Total No. of flowering plants	Mean No. of flowers per inflorescence	Mean inflorescence ht (cm)	Mean No. of fruit per inflorescence
1976 fire	Spring 1979	306	79·5±0·9	162·4±2·5	2·45±0·11
	Spring 1980	100	64·9±1·4	186·0±3·4	0·86±0·11
	Spring 1981	9	62·3±3·0	228·8±13·3	0·56±0·24
		Sample size			
1979 fire	Spring 1981	63	171·8±4·2	138·7±3·1	3·74±0·41

1979 there have also been decreases in the average number of flowers per inflorescence and in the average number of fruit per inflorescence and increases in the average inflorescence height. The differences between 1979 and 1980 are all significant ($P<0·05$, t-test); for 1980 and 1981 only the difference in mean inflorescence height is significant ($P<0·05$, t-test). During spring 1981 in the circles in the 1979 fire area (see Table 1), the inflorescence height averaged less than for any of the flowering periods in the 1976 fire area ($P<0·05$, t-test) while the numbers of flowers and fruits per inflorescence averaged more ($P<0·05$, t-test).

Flowering of Telopea *in Different Places*

During spring 1981 the density of flowering plants (i.e. number of flowering plants per circle) and the percentage of plants in flower declined with increases in the time since an area last burnt (Table 2). The differences in the percentage of flowering are significant between the 1979 and 1976 (or 1965) burn areas ($P<0·05$, t-test based on Normal approximation to binomial) but not between the 1965 and 1976 burn areas ($P>0·05$, t-test based on Normal approximation to binomial).

Flowering of Lambertia

During spring 1981 the peak density of *Lambertia* inflorescences declined with increases in the time since an area last burnt (Table 3). The differences between the areas are significantly different ($P<0·05$ for comparisons between 1979 and 1976 burn areas and between 1976 and 1965 burn areas, t-test).

For two circles in the 1976 burn area, the peak numbers of *Lambertia* inflorescences declined significantly from spring 1980 to spring 1981 ($P<0·05$, t-test) (Table 3).

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Discussion

There appear to be strong relationships between flowering and time since the last fire for both *Telopea* and *Lambertia*. In the case of *Telopea*, the percentage (or density) of plants in flower, the average number of flowers per inflorescence and the average number of fruits per inflorescence all peak 2 years after a summer fire and decline thereafter (Tables

Table 2. Flowering of *Telopea speciosissima* in different areas during spring 1981

Area	No. of circles	Mean No. of flowering plants per circle	Mean No. of plants per circle	Percentage of plants in flower
Ku-ring-gai Chase (1965 fire)	2	0·50	32·5±5·5	1·5 (n=65)
Brisbane Water (1976 fire)	4	0·75	138·8±19·4	0·5 (n=555)
Brisbane Water (1979 fire)	2	23·00	105·0±2·0	22·0 (n=210)

Table 3. Flowering of *Lambertia formosa* in different areas during spring 1980 and 1981

Area	Peak No. of inflorescences per circle					
	1980			1981		
	Mean	s.e.	n	Mean	s.e.	n
Ku-ring-gai Chase (1965 fire)	—	—	—	5·5	3·5	2
Brisbane Water (1976 fire)	195·5	17·5	2	96·5	7·8	4
Brisbane Water (1979 fire)	—	—	—	419·0	108·0	2

1 and 2). On the other hand, the average height of the inflorescences above the ground appears to increase with increases in time since the last fire (Table 1). This undoubtedly reflects the continual growth of the plants, starting from a rhizome, following a fire. The density of *Lambertia* inflorescences reaches a peak 2 or 3 years after a summer fire (Table 3). No information is currently available on either the cause or the adaptive significance of these relationships.

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